

3. Industry and Technology Landscape Analysis

3.1 Current State Assessment: As-Is Analysis

Faz Australia has their flagship product MedTalk that is an Artificial Intelligence medical scribe that automates clinical documentation during doctor and patient consultation using an ambient transcription. The MedTalk workflow automation generates the EMR (electronic medical record) ready reports that automatically updates the patient's record. MedTalk is connected to third-party software like Epic and Oracle Health Millennium with bidirectional feedback by EHR (electronic health record) integration. The EHR integration enables MedTalk to integrate with BPs (Best Practice like clinics or hospitals). MedTalk is built on strict data sovereignty standards from various frameworks like the ACSC Essential Eight, IRAP, HIPAA, GDPR AND APP (Australian Privacy Principles). Before the draft EMR updates the patient's record it is converted into structures formats using Australian medical coding protocol SNOMED CT-AU (Systematized Nomenclature of Medicine – Clinical Terms – Australian Extension) and ICD-10-AM (International Statistical Classification of Diseases and Related Health Problems, Tenth Revision, Australian Modification) using FHIR R5 framework (McBride, Lawley et al. 2012). FHIR R5 is Fast Health Interoperability Resources current release used as the interoperability standard (Lee, Mendoza et al. 2025). Literature reviews that successful implementation depends on the local configurations, workflow and mapping quality (Hornback, Marteau et al. 2025). All this is done using the MedTalk internal infrastructure. MedTalk partners with MediBetter BOSS (Billing Optimisation and Support Software) to map the ICD-10-AM format to Medicare Benefits Schedule (MediBetter, 2026).

Faz Australia utilises various testing tools, frameworks and platforms to test, map and synchronise data to the external system format it is integrating with. For data exchange integration, developers use SMART on FHIR that ensures that it can integrate with other BPs and adds a security authorisation layer OAuth2.0 to secure logins and user permissions. Inferno AU core testing kits is used to test the mapping done internally by MedTalk to ensure correct mapping between SNOMED CT-AU and ICD-10-AM (Hornback, Marteau et al. 2025). The developing platform utilised by Faz Australia is Epic sandbox that test the integration (SMART on FHIR) using synthetic data before actual patient data exchange. Epic sandbox is a comprehensive test environment and is costly to setup for commercial deployment. According to Choudhary (2026), charging fees are categorised into three parts, namely connection hub fees to load application approximately USD\$500. Vendor memberships services costs US\$1 900 and build labour ranging from USD\$40, 000 (Choudhary, 2026). Medtalk partnership with MediBetter costs is not publicly available but MediBetter do charge up to AUD\$500 flat fee (MarkiTech.AI, 2025). The major pain point for MedTalk is the upfront cost of the bidirectional FHIR integration with Epic and with Oracle Health Millennium which costs between AUD\$15 000 and AUD\$60, 000 (Software Finder, 2026).

The cost and limited flexible access to an existing sandbox create operational inefficiency and reduce the ability to test integrations repeatedly across different environments. The restricted access to sandboxes slows continuous deployment and integration. Moreover, Medtalk strong privacy and compliance requirements reduce testing flexibility in testing live connections with external systems. These limitations pose high pain points for Faz Australia as it costs more to expand, delayed deployments and non-compliance result in legal penalties. These limitations affect the developers' efficiencies and releases, which leads Faz Australia design this white label test harness to ensure developers have a reusable, compliant test harness. This strengthens the case for a reusable synthetic test harness.

3.2 Technology and Market Survey

MedTalk is an intelligent AI (Artificial Intelligence) healthcare platform for clinical documentation and integration. This health-tech industry has been strictly highly regulated by various laws, regulations and policy framework designed to protect sensitive patient information at rest and in transit. Therefore, the main challenge is not only generating accurate clinical notes but protecting the data exchange and correct translation to SNOMED CT-AU and ICD-10-AM. This technology survey will focus on Australian relevant technologies and frameworks as in healthcare there are various tools. The tools and frameworks used are Australian terminologies (SNOMED CT-AU, ICD-10-AM, MBS), HL7, FHIR, SMART on FHIR, Synthea and Halo connect.

Clinical terminology is essential and Australian extensions include SNOMED CT-AU and ICD-10-AM. SNOMED CT-AU is an Australian extension of global SNOMED CT, that is a vocabulary of medical terms tailored to the Australian Medicines Terminology (AMT) (McBride, Lawley et al. 2012). SNOMED CT-AU is used to record detailed patient health information consistently across Australian Healthcare Systems. The patient information is organised into hierarchies which include structures, procedures and clinical findings. The SNOMED CT-AU coding is then mapped to ICD-10-AM which classifies diseases and diagnosis according to Australian to Australia for health tracking and billing purposes. The ICD-10-AM is mapped to MBS Medicare benefits and help finalise billing invoice for patient. These terminologies are open-source product software standards that are required to be encoded within patient records to validate them (National Clinical Terminology Service, 2023).

Patient data to be shared from one system to another it first needs to be structured to a format that each system would be able to interpret. FHIR (Fast Healthcare Interoperability Resources) is the modern standard web-based technology used. FHIR is based on XML (extensible markup language) with an HTTP based RESTful (representational state transfer) protocol (Ayaz et al., 2021). The RESTful APIs (application programming interfaces) use HTTP operations to create resources, read and search enabling real time data exchange. FHIR resources are building blocks structure that includes patient ID, observations, practitioner and condition. The Australian Health Level Seven (HL7) perform the same function but is an older model. Moreover, the HL7 is a registered trademark that cannot be used without permission while FHIR is an open-source standard.

As mentioned in previous section MedTalk requires integration with BPs to allow exchange of patient information. MedTalk uses SMART on FHIR framework that allow systems to safely connect to BPs and share EHR. The SMART on FHIR allow interoperability assisting the receiving system to view the resources sent and transform to the receiving format. It then also adds a security layer OAuth 2.0 that authenticates user, authorise permission to use the resources. Another tool used is Halo Connect that connects local medical practices to the cloud-based software. Halo connect has two components the Halo cloud and Halo link. Halo cloud is used to connect to local database and transform SQL formats to FHIR resources transported to the external system using the Halo link. Both integration methods work but as per the test harness project Halo connect is most suited as the mock would be stored in the database.

Synthetic patient data is required to maintain compliance. Synthea is the most credited mock patient generator as it generates a complete lifespan from birth to death of patient health records. (Walonoski, Hall et al. 2022). The major limitation for Synthea is that it creates localised synthetic data using demographics from United States of American since the project require Australian clinical patterns and terminology. It can be customised by adding demographic files

of the country you would like for example Australia. CSIRO customised Australian mock data for Queensland and the mock patient data are available publicly (Diouf, Grimes et al. 2024). The major limitations is there are no publicly available Australian customised Synthea to generate as per user need. The test harness will leverage the Australian synthetic data as some tools requires real patient data to create synthetic data.

Above summaries the Australian frameworks, tools and standards required to mimic a healthcare system. The main gap is developing a test harness that leverages the standards which define how data should be exchanged, integration, authentication and terminology define how meaning should be coded. Then perform tests to validate integration, terminology and mapping to ensure that data being exchanged is accurate and complete.

3.3 Comparative Analysis of Existing Solutions

The table below compares existing sandbox that provides valuable environment to test interoperability of the health care systems. SMART Health IT Sandbox only focuses on validating the security layer of the EHR exchange between systems while Epic on Epic App Orchard is comprehensive and requires the application to be load on and expensive upfront cost. MediRecords Developer Sandbox provide developing environment and only perform integration with MediRecords resources not any external systems. Synapxe HealthX innovation sandbox main limitation is that its supports Singapore healthcare instead of Australian. Inferno Test Kit AU Core only validates mapping. The proposed test harness addresses a gap in existing solutions by combining synthetic data, interoperability testing and terminology validation in a single reusable environment. Its value lies in repeatability, safety and support for Australian clinical workflows.

Table 3.3 Comparative analysis of existing solutions

Technology	Core Functionality	Deployment Model	Limitations	Relevance to project
SMART Health IT Sandbox	Provides simulated EHR environ for testing HL7 FHIR APIs and OAuth 2.0 authentication and patient data exchange (SMART on FHIR, n.d.).	Cloud based sandbox	Testing in the sandbox does not guarantee compatibility with HER implementation	Partly relevant test FHIR interoperability and SMART on FHIR.
Epic on Epic App Orchard	Developer ecosystem for building and distributing applications that integrate with Epic EHR. Provides APIs based on HL7	Cloud/EHR integrated	Integration can involve significant development, certification and commercial costs. Access is to production data	Relevant but does not validate the terminology.

	FHIR, testing environments, documentation and integration capabilities (Bonami Software, 2026).		and APIs is restricted.	
MediRecords Developer Sandbox	Provides developers with an environment to test applications and integrations with MediRecords (MediRecords Connect, 2026)	Cloud based developer sandbox	Sandbox data and functionality may differ from the production environment.	Moderately Relevant but does not validate the terminology.
AU Core Inferno Test Kit	Is a web-based testing tool used to validate FHIR's server API endpoints, Australian core profile (SNOMED CT-AU)(Hornback, Marteau et al. 2025)	Web- based	Only validates server endpoint to Australian standards and Australian Core profiles.	Relevant but does not test integration.
Synapxe HealthX innovation sandbox	Provides an innovation and experimentation environment for healthcare technologies, supporting testing, prototyping and validation of digital health solutions and interoperability concepts within a healthcare ecosystem (Synapxe, 2023)	Innovation sandbox/ testing environment	Primarily focused on Singapore's healthcare ecosystem different from Australian.	Moderately relevant as a reference model for testing healthcare technologies

3.4 Gap Analysis: As-Is to To-Be

The main challenge the test harness would address is having a reusable full synthetic test environment fully accessible to developers. Currently Faz Australia reply on various partners which is costly, have restricted access and delay updates. This solution closes this gap and creates an independent platform that can be utilised by various systems.

The second gap manual clinical notes verification as to validated encoded clinical notes generated by MedTalk. Currently Medtalk supports structured output that encode the clinical notes using SNOMED CT-AU. There are not published mechanisms that claim that MedTalk can verify and detect semantics that are mismatched or dropped when data is exchanged. The proposed dataset will close this gap by validating the completeness of the data being exchanged.

The third gap is endpoint API connectivity as to standard aligned true interoperability. A BP system can receive or send data but ignore or fail to meet Australian standards like SNOMED CT-AU or FHIR or ICD-10-AM. This can cause a critical clinical risk like rejected messages, incorrect information shared and administrative failures. The proposed test harness tackles this challenge by validating what MedTalk sends and receives implementing the Australian standards making integration behaviour testable and reusable.

The fourth gap is sunny day testing as to resilient testing. In day-to-day integrations fail because of delays, malformed payload, dropped connections and other edge cases. Before deployment of new releases of MedTalk all failure conditions should be tested otherwise it would behave unpredictably when connected to external systems. The proposed test harness would provide a testing environment that completes repeatable robust error simulations like slow, bad data and connection loss.

The final gap is fragmented tooling as to single validation workflow that validates terminology mapping, integration and bad connection. The unified controlled environment would reduce fragmented testing complexity and reduce inconsistent validation processes. The proposed test harness closes this gap through merging all the standards and validate them through automated testing.

3.5 Justification of Proposed Solution

Faz Australia Medtalk mappings from SNOMED CT-AU to ICD-10-AM and Australian profiles are currently being validated by external tool AU Core Inferno Testing kit. This testing kit does not provide a repeatable complete end to end assurance process of FHIR interoperability and coding translation across real integration configuration before production. As Medtalk evolves, developers still rely on manual review of FHIR payloads in testing environments. This introduces clinical risks like malformed payloads, incorrect connector updates and compliance risk due to limited evidence that mappings and failure -handling remain correct overtime. Therefore, the proposed Test harness builds on existing AU Core standards with a fully synthetic isolated FHIR testing environment. It enables safe reusable validation of integration behaviour, payload integrity, terminology mapping and simulated failure modes. This project strengthens MedTalk's cybersecurity posture and operational efficiency by providing a reusable testing environment that reduce slow deployments, defects, privacy breaches and coding noncompliance when onboarding new EMR integrations with other BPs.

Reference

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